

# AKSHAY BAIPLAWAT

Software Engineer | Systems Programming & AI Orchestration | On-Device LLM Infrastructure

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## Education

MOTILAL NEHRU NATIONAL INSTITUTE OF TECHNOLOGY, ALLAHABAD , INDIA

2019 - 2023

BACHELOR OF TECHNOLOGY

## Skills

### AI & Agentic Development:

- **AI Engineering:** Development utilizing Claude Code, Prompt Engineering, Multi-agent Pipeline Design, Retrieval-Augmented Generation (RAG), Semantic Search (Vector Databases) and Multimodal Context Injection.
- **On-Device AI:** LiteRT (GPU), llama.cpp (JNI), MediaPipe and GGUF/LiteRT model optimization

### Core Backend & Systems

- **Languages:** C++ (Systems-level), Core Java, TypeScript, SQL, Groovy and Apex.
- **Frameworks:** SAP Integration Suite, Spring Boot, React, Kibana (OctoBus), CMake and CI/CD (GitHub Actions).
- **Fundamentals:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, Clean Architecture and Test-Driven Development (TDD).
- **Systems Programming:** Memory Management, Multi-threading, Windows Native (Win32) Development, Concurrency Models (Binary CAS Mutexes).

## Work Experience

SAP LABS INDIA (Concur Integration) | Software Engineer

Nov' 24 - Present

- **Architected a deterministic traffic simulation engine** using custom Mock APIs to stress-test high scale integrations, identifying critical JMS Queue and memory bottlenecks that would have blocked onboarding for 1,000+ enterprise clients.
- **Engineered an automated transport engine** reduced artifact transport time latency by 99% (from 5 minutes to seconds), reclaiming 50+ monthly engineering hours for the internal engineering team.
- **Technical Lead for the Salesforce Connector V2 migration**, spearheading the end-to-end modernization of legacy APIs to a Second-Generation Managed Packaging (2GP) model while resolving critical customer-facing defects ahead of the Q3 deadline.
- **Optimized system observability and SLOs** as Incident Manager (2025 Jan - 2026 Jan) by overhauling PagerDuty alert logic and logging dashboards, significantly accelerating "On-Call" response times and production incident resolution.
- **Designated Product Line Owner for the ADP line (WorkforceNow, WorkforceNow NextGen, Run, Vantage)**, orchestrating high-scale service updates and maintaining 100% alignment with release calendars to guarantee zero-downtime for global enterprise clients.
- **Executed a dual-track Development and Quality Engineering strategy**, personally identifying and neutralizing two critical production defects that threatened the entire Concur Professional NetSuite user base.

ZOHO Corporation | Software Developer – Systems & Low-Level Programming

Aug' 23 - Nov' 24

- **Optimized session initialization flows** for a remote access agent supporting 1M+ daily sessions, achieving a 30% reduction in latency and a 15% decrease in executable size.
- **Engineered the Zoho Assist Windows Agent in C++**, ensuring 100% compatibility across all legacy and modern Windows OS versions for a global remote-access user base.

- **Neutralized a critical, industry wide cursor blocking defect** by manipulating low level cursor input signal, positioning Zoho Assist as the first remote support software to operate without the flaw.
- **Developed high performance C++ wrappers** for third-party libraries (POCO, rapidJSON), resulting in a 15% performance gain in core agent functions through optimized memory and serialization patterns.
- **Implemented a secure session audit reporting engine**, enhancing customer security metrics by 35% through detailed session logging and data transparency.
- **Facilitated cross-functional integrations** by resolving complex architectural bottlenecks, enabling the successful embedding of Zoho Assist into multiple partner product lines.

## Technical Research & Projects

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### Core AI: On-Device LLM Platform | Architect (Utilizing Gemini and Claude Code) [Link](#)

- **Architected a multi-tenant on-device LLM inference platform** using a custom AIDL service contract for cross-process communication, enabling third-party applications to execute sub-second inference with zero cloud dependency.
- **Engineered a pluggable ML backend** utilizing LiteRT (GPU), llama.cpp (JNI), and MediaPipe to support GGUF/LiteRT models; optimized template resolution by 99% through zero-copy metadata parsing O(H) .
- **Optimized Time-to-First-Token (TTFT)** via one-way AIDL callback streaming and implemented a 4-tier chat template resolver for Llama, Gemma, and Mistral model families
- **Designed a three-tier concurrency model** using Binary CAS Mutexes and single-threaded JNI dispatchers to prevent data races across N concurrent client applications.

### Finance Tracker |UI, Engineer & Architect (Utilizing Gemini and Claude Code) [Link](#)

- **Architected a privacy-centric Android application** using MVVM and Clean Architecture with Jetpack Compose and Kotlin Flow, ensuring 100% reactive data streams and zero-cloud-dependency.
- **Engineered an intelligent SMS Parsing Engine** using regex-based pattern matching and merchant normalization to automate transaction categorization for major Indian banks (HDFC, ICICI, SBI).
- **Optimized group debt resolution** by implementing a Greedy Settlement Algorithm, reducing potential transaction complexity from  $O(N^2)$  to  $O(N)$  and validating logic through 23+ targeted unit tests.

### JARVIS: Multimodal AI Voice Assistant | AI Engineer [Link](#)

- **Architected a real-time multimodal agent** using LiveKit Agents SDK and WebRTC, orchestrating a low-latency STT → LLM → TTS pipeline with <1s perceived response time through Voice Activity Detection (VAD).
- **Engineered a VisionAssistant class** to process asynchronous video streams, injecting real-time visual context into Gemini 2.5 Flash for vision-grounded reasoning and situational awareness.
- **Orchestrated complex productivity workflows** via the Model Context Protocol (MCP), allowing the LLM to autonomously manage Google Calendar and Gmail operations through secure tool-calling patterns.
- **Developed a Test-Driven Development (TDD) framework** utilizing an LLM-as-judge evaluation pattern to validate grounding, intent, and safety protocols across Python code.